

BTK: 2 molecules stopped early, all small molecules

5 stopped trials, but 2 molecules — all small molecules: Ibrutinib and Vocabrutinib. Against 5.0% for all oncology trials with a resolved drug target and 4.3% across all oncology Phase 2,3 trials in the same window. Rates count trials that stopped early for efficacy, safety or benefit–risk reasons; trials that completed and missed their endpoints are not counted.

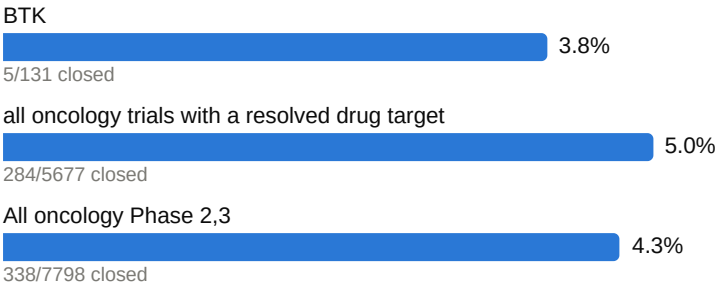
3.8%
5 of 131 closed trials (95% CI 1.6%–8.6%)

4 / 1 / 0
efficacy / safety / both (adds to 5)

32.7%
of 401 trials have closed · 270 still open or unresolved

0.9%
stopped for a biological reason within 3 years of starting (95% CI 0.3%–2.9%; 227 still at risk; 1.5% area-wide)

How this compares



What is in this class: 138 distinct experimental drugs across 401 trials (small molecule 396, antibody 184, unknown 40, adc 10). Most tested: IBRUTINIB (155), ACALABRUTINIB (85), RITUXIMAB (83), ZANUBRUTINIB (74), VENETOCLAX (60). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 5 stops came from 5 sponsor-asset programmes across 5 sponsors; the largest (Sunesis Pharmaceuticals) contributed 1; removing that programme's trials from both sides leaves 3.1%. Screened alongside 83 other segments in this area, it does not survive a 10% false-discovery correction (q=1, Benjamini-Yekutieli). A further 7 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 9.2%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	4	77	5.2%	2.0%–12.6%
2018–2020	1	40	2.5%	0.4%–12.9%
2021–2024	0	14	0.0%	0.0%–21.5%

Where the stops sit

Efficacy / futility: 4, Safety: 1.
Sponsors with most stops: Johnson & Johnson (1), National Heart, Lung, and Blood Institute (NHLBI) (1), Sunesis Pharmaceuticals (1), AbbVie (1), Fondazione Italiana Linfomi - ETS (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02703272	3	Johnson & Johnson	IBRUTINIB, RITUXIMAB, IFOSFAMIDE, CARBOPLATIN	2021-06	Efficacy	IDMC recommended that enrolment be stopped, as the EFS hazard ratio and associated p-value crossed the futility boundary specified in protocol (July 2020).
NCT04209621	2	National Heart, Lung, and Blood Institute (NHLBI)	DUVELISIB, IBRUTINIB	2021-01	Safety	Due to unexpected sudden death on study.
NCT03037645	2	Sunesis Pharmaceuticals	VECABRUTINIB	2020-08	Efficacy	Phase 1b portion completed. Sponsor decided not to proceed with P2 portion of study. Vocabrutinib was very well tolerated, there was insufficient evidence of activity at the doses tested in the Phase 1b to advance to Phase 2.
NCT02351037	2	AbbVie	IBRUTINIB, LD-AraC, AZACITIDINE	2017-04	Efficacy	Study data do not support development in AML.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02670317	2	Fondazione Italiana Linfomi - ETS	OBINUTUZUMAB, IBRUTINIB, CYCLOPHOSPHAMIDE ANHYDROUS, DOXORUBICIN	2017-02	Efficacy	GA101-CHOP not advantage from rituximab-CHOP

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: trials that completed with negative results are not counted, and programs discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 32.7% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

Any mechanism, sponsor or indication, updated weekly. The dataset behind this brief covers every stopped oncology trial with an efficacy or safety signal plus the full denominator universe. Free sample and licensing: clinicaltrialfailures.com/data-licensing

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.